

Data Analytics Insights Report

Project: Smart City Traffic Analytics Capstone

1. Yearly Traffic Trends

Based on our SQL database analysis, total traffic volume peaked in 2017 with over 35.4 million vehicles recorded across 10,605 logged hours. Year 2015 showed a lower recorded volume (13.8 million vehicles across 4,373 hours) due to fewer data collection timestamps in the dataset.

2. Descriptive Statistics

- **Mean Traffic Volume:** 3,259.82 vehicles per hour
- **Median Traffic Volume:** 3,380.00 vehicles per hour
- **Standard Deviation:** 1,986.86 vehicles per hour
- **Variance:** 3,947,613.88
- **Range:** 7,280 vehicles (0 to 7,280)

3. Probability & Congestion Analysis

- $P(\text{Congestion} > 5,500 \text{ vehicles}) = 14.73\%$
- $P(\text{Clear Weather}) = 27.65\%$
- $P(\text{Clear Weather} \mid \text{Congestion}) = 20.32\%$

Key Insight: Weather conditions and traffic congestion are **not independent**. Drivers experience clear weather during heavy congestion less frequently (20.32%) than in general conditions (27.65%), confirming that cloud cover and minor precipitation coincide with higher peak-hour demand.